

Answer Key — Mathematics Grade 10

Sub-Strand 3.1: Trigonometry I • Lesson 4: Tangent and the Sine–Cosine–Tangent Relationship:
Solving the Mango-Tree Problem

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
D	A	B	B	C	D	C	B	B	A

USE AFTER: PREDICT PHASE — Predict: silent prediction and think-pair-share on which ratio is most useful for the mango-tree height

Q1. In Kisumu, a student measures the sun's angle with the ground as 53° . The mango tree's shadow on the ground is 9.0 m long. She wants the tree's height. The tree is vertical and the ground is horizontal. Which ratio links the two lengths she knows or wants (height and shadow) with the 53° angle?

- A. No ratio works unless she first measures the length of the sun's ray (the hypotenuse)
- B. $\sin 53^\circ$, because sine is the most useful ratio for every triangle
- C. $\cos 53^\circ$, because the shadow is on the ground
- D. $\tan 53^\circ$, because it links the opposite side (height) and the adjacent side (shadow)**

Correct answer: D. The height is opposite the 53° angle and the shadow is adjacent to it. Tangent is Opposite/Adjacent, so it connects these two sides without needing the hypotenuse.

USE AFTER: OBSERVE PHASE — Part A: Algebraic Derivation of $\tan \theta = \sin \theta / \cos \theta$

Q2. In a right-angled triangle, $\sin \theta = O/H$ and $\cos \theta = A/H$. What do we get when we simplify $(O/H) \div (A/H) = (O/H) \times (H/A)$?

- A. O/A , which is $\tan \theta$**
- B. O/H
- C. A/O
- D. H/A

Correct answer: A. The H in the numerator cancels with the H in the denominator, leaving O/A . This is Opposite/Adjacent, the definition of $\tan \theta$.

USE AFTER: OBSERVE PHASE — Part A: Algebraic Derivation, where $\tan \theta = \text{Opposite} / \text{Adjacent}$ (TAO) is written

Q3. A right-angled triangle has sides of 6 cm, 8 cm and 10 cm (the hypotenuse). For the angle θ opposite the 6 cm side, what is $\tan \theta$?

- A. 1.33
- B. 0.75**
- C. 0.6
- D. 0.8

Correct answer: B. $\tan \theta = \text{Opposite/Adjacent} = 6/8 = 0.75$. The value 0.6 is $\sin \theta$ (6/10), 0.8 is $\cos \theta$ (8/10), and 1.33 is the inverted ratio 8/6.

USE AFTER: OBSERVE PHASE — Part B: Numerical Verification table (40° row)

Q4. From mathematical tables, $\sin 40^\circ = 0.6428$ and $\cos 40^\circ = 0.7660$. Using $\tan \theta = \sin \theta \div \cos \theta$, what is $\tan 40^\circ$ to 3 decimal places?

- A. 1.409
- B. 0.839**
- C. 1.192
- D. 0.492

Correct answer: B. $\tan 40^\circ = \sin 40^\circ \div \cos 40^\circ = 0.6428 \div 0.7660 \approx 0.839$. The value 0.492 comes from multiplying, 1.192 from dividing cos by sin, and 1.409 from adding.

USE AFTER: OBSERVE PHASE — Part B: Numerical Verification, comparing column 4 and column 5

Q5. A pair finds that for one angle, $\sin \theta \div \cos \theta$ is 0.8391 and $\tan \theta$ read from tables is 0.8390. What is the most sensible explanation of this tiny difference?

- A. They used the wrong triangle, because $\tan$ changes with the size of the triangle
- B. The identity only works for 45°
- C. Table values are rounded, so tiny differences are expected**
- D. The identity $\tan \theta = \sin \theta / \cos \theta$ does not work for that angle

Correct answer: C. Table values are rounded to a few decimal places, so tiny differences are normal. The identity itself is exact for every acute angle.

USE AFTER: EXPLAIN PHASE — Part A: Sensemaking Discussion, Q1 (is $\tan$ unnecessary?)

Q6. A learner says, "We can already get $\tan \theta$ from $\sin \theta$ and $\cos \theta$, so $\tan$ is unnecessary." What is the best reply?

- A. $\tan$ can only be used when the hypotenuse is known
- B. $\tan$ is only used for angles bigger than 90°
- C. They are right: $\tan$ is redundant and is never needed
- D. $\tan$ is a convenient shortcut when we know an angle and one side but not the hypotenuse**

Correct answer: D. $\tan$ is not independent of $\sin$ and $\cos$, but it is convenient. It connects the opposite and adjacent sides directly, without needing the hypotenuse.

USE AFTER: EXPLAIN PHASE — Part A: Sensemaking Discussion, Q2 (which side is 9.0 m: opposite or adjacent?)

Q7. The sun makes an angle of 53° with the ground. A vertical mango tree of height h casts a 9.0 m shadow along the ground, and the 53° angle is at the end of the shadow, between the ground and the sun's ray. Which statement is correct?

- A. The 9.0 m shadow is the side opposite the 53° angle, and the tree height h is the side adjacent to it
- B. Both the shadow and the tree are adjacent to the 53° angle
- C. The 9.0 m shadow is the side adjacent to the 53° angle, and the tree height h is the side opposite it**
- D. The 9.0 m shadow is the hypotenuse

Correct answer: C. The shadow lies along the ground next to the 53° angle, so it is adjacent. The tree stands across from the angle, so it is opposite. The hypotenuse is the sun's ray.

USE AFTER: EXPLAIN PHASE — Part B: Mango-Tree Calculation ($h = 9.0 \times \tan 53^\circ$)

Q8. A mango tree has a shadow 9.0 m long when the sun makes an angle of 53° with the ground. Given $\tan 53^\circ = 1.3270$ and $\tan 53^\circ = h \div 9.0$, what is the height h of the tree?

- A. 5.42 m
- B. 11.943 m**
- C. 7.19 m
- D. 6.78 m

Correct answer: B. From $\tan 53^\circ = h/9.0$, $h = 9.0 \times \tan 53^\circ = 9.0 \times 1.3270 = 11.943$ m. The value 6.78 m comes from the inverted ratio $9.0 \div 1.3270$.

USE AFTER: EXPLAIN PHASE — Part B: Mango-Tree Calculation, shadow-ratio cross-check

Q9. A student is 1.6 m tall and her shadow is 1.2 m long. At the same time, a mango tree has a shadow 9.0 m long. Using similar triangles, $(\text{tree height}) \div 9.0 = 1.6 \div 1.2$. What is the height of the tree?

- A. 14.4 m
- B. 12.0 m**
- C. 9.4 m
- D. 6.75 m

Correct answer: B. $h = 9.0 \times (1.6 \div 1.2) = 9.0 \times 1.333 = 12.0$ m. This agrees with the tangent method (about 11.94 m) to within rounding.

USE AFTER: MODEL BUILDING PHASE — Trigonometry Model: adding the 'Ratio Web' annotation ($\tan \theta = \sin \theta / \cos \theta$)

Q10. On the 'Ratio Web' in the Trigonometry Model, $\tan \theta$ is linked to $\sin \theta$ and $\cos \theta$. For an angle of 53° , $\sin 53^\circ \approx 0.799$ and $\cos 53^\circ \approx 0.602$. Using the identity, what is $\tan 53^\circ$ to 3 decimal places?

- A. 1.327**
- B. 0.481
- C. 0.753
- D. 1.401

Correct answer: A. $\tan \theta = \sin \theta \div \cos \theta = 0.799 \div 0.602 \approx 1.327$. Dividing cos by sin gives 0.753, which reverses the identity, and multiplying gives 0.481.